Evaluating the Credibility of the FFA Face-Selectivity Literature: A Z-Curve Analysis

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Course: PSYC 6018 · Instructor: Rick Thomas · May 5, 2026

Abstract

Concerns about publication bias, questionable research practices, and the replication crisis have prompted calls for systematic credibility audits of established research literatures. We applied Z-curve 2.0 to a preregistered sample of 25 significant focal z-scores drawn from 27 published fMRI studies reporting the FFA face-selectivity contrast (faces > non-face objects in the fusiform face area), in order to estimate the expected replicability and evidential value of this literature. Both preregistered predictions were confirmed. We find that the Expected Discovery Rate (EDR = 0.86, 95 % CI $[0.80, 0.98]$) substantially exceeded the 5 % nominal alpha level, and the Expected Replication Rate (ERR = 0.86, 95 % CI $[0.82, 0.96]$) was well above zero. The Expected False Discovery Rate was negligible (EFDR = 0.009). Observed and Expected Discovery Rates were nearly identical, and the implied file-drawer ratio was low (= 0.17), suggesting that the published record is not substantially distorted by selective reporting. We interpret these findings as evidence that the FFA face-selectivity contrast has genuine and replicable evidential value. Substantive interpretation of EDR is conditioned on two model stability warnings due to small-sample fit and single-component weight concentration, which we document in full. Implications for open science practice in human neuroimaging are discussed.

Keywords: Z-curve, publication bias, replication, fusiform face area, face selectivity, credibility analysis, fMRI

Introduction

Publication bias, questionable research practices (QRPs), and failures to replicate have raised serious concerns about the credibility of published scientific findings (Open Science Collaboration, 2015). Journals disproportionately publish statistically significant results, and researchers may, consciously or not, make analytic decisions that inflate the evidence in favor of a hypothesis (Fiedler & Schwarz, 2016). Together these forces can produce bodies of literature in which the Observed Discovery Rate (ODR) far exceeds the rate that would obtain if all conducted tests were reported, and in which effect-size estimates are systematically inflated. Z-curve analysis addresses this directly by modeling the distribution of significant test statistics as a finite mixture of noncentral normals and extrapolating into the non-significant region to estimate the Expected Discovery Rate (EDR) and Expected Replication Rate (ERR) corrected for selective reporting (Bartoš & Schimmack, 2022; Brunner & Schimmack, 2020).

We applied Z-curve analysis to the fMRI literature on face selectivity of the fusiform face area (FFA), a canonical finding in cognitive neuroscience. Since Kanwisher et al. (1997) first identified the FFA as preferentially responsive to faces, numerous studies have reported stronger FFA activation for faces than for non-face objects (Kanwisher & Yovel, 2006). A competing visual-expertise account holds that FFA reflects high-level perceptual expertise rather than face-specificity per se (Gauthier et al., 1999). Regardless of this theoretical debate, the statistical credibility of the basic face-selectivity contrast (faces > objects in FFA) has not been systematically evaluated. We asked whether the published record reflect genuine, replicable signal, or could it be substantially inflated by selective reporting and low statistical power? A well-powered literature should yield high ERR and EDR with a small gap between observed and expected discovery rates; a literature distorted by selective reporting should produce the opposite.

Preregistration Summary

This study was preregistered on AsPredicted (ID: mp88kc35ps7hz8ef9ad8ny5bj374p72925) prior to any data collection. All key decisions—research question, inclusion and exclusion criteria, extraction procedures, statistical conversion rules, and analytic approach—were specified in advance. We preregistered two directional predictions. Namely, (1) the EDR would exceed the nominal $\alpha = .05$ level, and (2) the ERR would be substantially above zero. An EDR near 5% combined with a near-zero ERR would have been interpreted as evidence that the literature lacks genuine evidential value. The focal dependent variable was one confirmatory inferential test per independent sample for the Faces > Non-Faces contrast in FFA; descriptive statistics and exploratory analyses were excluded. Z-curve analysis was preregistered to be conducted using the R package `zcurve` (Bartoš & Schimmack, 2022), with two independent coders extracting data and discrepancies resolved by adjudication. The study was conducted in full accordance with the preregistered protocol; no deviations occurred.

Method

Literature Domain, Eligibility, and Source Selection

We conducted a Z-curve credibility audit of the fMRI literature on FFA face selectivity. Thirty candidate articles were identified from the published literature with no restriction on publication year; all materials are publicly archived at https://github.com/Mainakdeb/research-design. Eligibility decisions were preregistered prior to data extraction. Studies were included if they used task-based fMRI with human neurotypical participants, reported a univariate BOLD contrast of faces > non-face objects localized to FFA (defined functionally or anatomically), and provided a p-value or sufficient statistics (t, F, χ^2, or r with appropriate degrees of freedom) to compute an exact two-sided p-value. Studies using non-fMRI modalities, clinical or non-human samples, or reporting only resting-state, connectivity, or multivariate analyses were excluded, as were manipulation checks, secondary outcomes, and exploratory analyses. The focal hypothesis test for each included study was the confirmatory univariate contrast of greater BOLD signal in FFA for faces than for non-face objects. When multiple eligible contrasts appeared in one article, preregistered tie-breaking rules were applied. To be more specific, the left-hemisphere statistic was preferred for bilateral reports, and the first confirmatory comparison was retained when multiple contrasts were available.

Data Extraction and Inter-Coder Reliability

Two coders independently extracted each study's design, independent and dependent variables, focal test statistic with degrees of freedom, reported p-value, and ambiguity notes. A third coder served as adjudicator for contested cases. Initial percent agreement was 83.3% (25/30 articles). The five disagreements reflected ambiguity about whether an article reported a usable focal FFA contrast. After adjudication, three articles (identifiers 19, 26, 27) were excluded for insufficient statistical detail, yielding 27 articles with extractable focal tests. Coding rules were not modified after adjudication.

Statistical Conversion and Z-Curve Analysis

All statistics were converted to two-sided p-values using the appropriate R distribution functions (`pt`, `pf`, `pchisq`, `pnorm`); one-tailed p-values were doubled before conversion. Inequality-reported p-values (e.g., $p < .05$) were coded conservatively as the threshold. Each p-value was converted to an absolute z-score via $|z| = \Phi^{-1}(1 - p/2)$. The 27 z-scores were then submitted to Z-curve 2.0 (Bartoš & Schimmack, 2022; Brunner & Schimmack, 2020) via the `zcurve` R package, which fits a finite mixture of seven normal components (means fixed at $\mu = 0$–6, unit SD) to z-scores exceeding $z = 1.96$, and extrapolates the implied population density into the non-significant region to estimate the EDR. The ERR, EDR, EFDR, file drawer ratio, and 95 % confidence intervals (1,000 bootstrap resamples) were obtained from this fit.

Results

Of the 30 articles entered into the extraction pool, three were excluded as ambiguous (identifiers 19, 26, and 27), yielding 27 articles with extractable focal z-scores. Of these, 25 exceeded the significance threshold ($z > 1.96$) and contributed to the Z-curve fit while the remaining two were borderline ($z = 1.96$, $p = .05$) and treated as non-significant. The ODR over the included sample was 0.93 (25/27, 95 % CI $[0.74, 0.99]$) and the corresponding rate over all 30 candidate articles was 0.83.

Focal z-scores ranged from 1.96 to 4.89, with the bulk concentrated between $z = 2.7$ and $z = 3.9$ (median = 3.29, modal value ≈ 3.89). The absence of a spike immediately above $z = 1.96$ is inconsistent with p-hacking (Simonsohn et al., 2014). Table 1 reports the primary Z-curve 2.0 outputs and Figure 1 displays the model fit. The ERR was 0.86 (95 % CI $[0.82, 0.96]$), implying that direct replications would be expected to reach significance ≈86 % of the time, which was well above conventional power benchmarks (Cohen, 1988). The EDR (= 0.86, 95 % CI $[0.80, 0.98]$) closely matched the ODR, and the File Drawer Ratio was low (= 0.17, 95 % CI $[0.02, 0.25]$), implying only ≈2 missing null results. The EFDR was negligible (= 0.009, 95 % CI $[0.001, 0.013]$). Inspection of the fitted mixture weights revealed that nearly all model mass concentrated on a single component at $\mu = 3$ ($w = 0.96$, 95 % CI $[0.37, 1.00]$), indicating a homogeneous, high-power generative process.

The `zcurve` package issued two automated stability warnings: (1) the input sample of $n = 25$ significant z-scores is small relative to corpus sizes for which Z-curve 2.0 was developed, which can produce biased EDR estimates and undercoverage of confidence intervals; and (2) placing $> 90\%$ of model weight on a single component makes the EDR especially sensitive to extrapolation assumptions. Both warnings are reflected in the wide bootstrap CI on EDR (interval width = 0.18). We treat ERR, which is anchored in the observed significant z-scores rather than extrapolation, as the primary estimate and interpret EDR with corresponding caution.

Discussion

Both preregistered predictions were confirmed. The EDR (= 0.86) substantially exceeded the 5 % nominal alpha level, and the ERR (= 0.86) was well above zero. When EDR ≈ ODR, as observed here, the model implies that the proportion of significant findings in the literature approximates what would be expected if all conducted tests were reported, with no large file drawer of suppressed nulls. The low File Drawer Ratio (= 0.17) and negligible EFDR (= 0.009) corroborate this interpretation. An ERR of 0.86 substantially exceeds the 50 % power threshold typically considered adequate (Cohen, 1988) and is inconsistent with the pattern of low ERR and inflated ODR characteristic of literatures heavily contaminated by publication bias (Brunner & Schimmack, 2020). The absence of a spike immediately above $z = 1.96$, a key signature of p-hacking (Simonsohn et al., 2014), further supports the integrity of the literature. Taken together, these results portray the FFA face-selectivity literature as unusually well-powered by the standards of human neuroimaging (Button et al., 2013), and suggest that researchers planning direct replications of the faces > objects contrast in FFA can treat an ERR of 0.86 as a reasonable prior on success probability. Importantly, these findings do not adjudicate between domain-specificity (Kanwisher & Yovel, 2006) and visual expertise accounts (Gauthier et al., 1999). Additionally, the high ERR confirms that studies reliably detect *something* in FFA, not that face-specificity is the correct theoretical interpretation.

Limitations

Several limitations qualify these conclusions. First, the small eligible sample ($n = 25$ significant z-scores) triggered a package-level stability warning as small-sample Z-curve fits can produce biased EDR estimates and confidence intervals with empirical coverage below 95 % (Bartoš & Schimmack, 2022). The EDR CI width of 0.18 reflects this instability, and EDR should be interpreted with corresponding caution. Second, concentration of model weight on a single mixture component ($w = 0.96$ at $\mu = 3$) means the EDR is estimated almost entirely by extrapolation into a region with no observed data, further limiting its precision. Third, the focal contrast definition (univariate BOLD for faces > objects in FFA) is narrower than some operationalizations in the literature, and formal sensitivity analyses varying inclusion criteria were not conducted. The three excluded articles (identifiers 19, 26, 27) were removed for insufficient statistical detail; their exclusion introduces unquantifiable uncertainty, mitigated by the fact that the rule was preregistered and applied symmetrically.

Finally, Z-curve addresses the statistical credibility of reported tests but cannot speak to effect-size replicability or construct validity (Cumming, 2014); it is best regarded as a complement to conventional meta-analysis rather than a substitute.

Conclusion

Z-curve analysis of 25 significant focal z-scores from the fMRI FFA face-selectivity literature returned high estimated average power (ERR = 0.86, 95 % CI [0.82, 0.96]) and a negligible expected false discovery rate (EFDR = 0.009), with the Estimated and Observed Discovery Rates in close agreement. The low File Drawer Ratio and the small implied count of missing null results are inconsistent with a literature substantially distorted by selective reporting. These findings support the conclusion that the FFA face-selectivity contrast has genuine and replicable evidential value, and stand in contrast to many other human neuroimaging literatures in which Z-curve analyses have revealed markedly lower power estimates (Button et al., 2013).

Substantive interpretation of the EDR point estimate should be conditioned on the model stability warnings documented in the Results; the ERR, which does not depend on model extrapolation, is the more robust primary estimate. More broadly, this project demonstrates the value of preregistered, transparent credibility audits as a complement to traditional meta-analysis and narrative review. Future work should pursue a more exhaustive literature search to enlarge the eligible sample, conduct formal sensitivity analyses varying inclusion criteria, and integrate the present credibility indicators with conventional effect-size meta-analysis. As the open science movement continues to reshape standards for transparency and reproducibility in cognitive neuroscience, the tools and practices exemplified here—preregistration, double coding, public data archiving, and distribution-based credibility analysis—offer a practical framework for evaluating the robustness of established empirical literatures.

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Table 1

Z-Curve 2.0 Outputs From 25 Significant Focal z-Scores in the FFA Face-Selectivity Literature

Metric	**Estimate**	**95 % CI**
Observed Discovery Rate (ODR)	0.93	[0.74, 0.99]
Expected Replication Rate (ERR)	0.86	[0.82, 0.96]
Expected Discovery Rate (EDR)	0.86	[0.80, 0.98]
Expected False Discovery Rate (EFDR)	0.009	[0.001, 0.013]
File Drawer Ratio	0.17	[0.02, 0.25]
Implied missing null results	2	[−1, 4]

Note. ODR computed as proportion of the 27 included focal z-scores exceeding $z = 1.96$. ERR, EDR, EFDR, File Drawer Ratio, and implied missing nulls estimated by Z-curve 2.0 (EM algorithm) on the 25 significant z-scores; 95 % CIs from 1,000 nonparametric bootstrap resamples.

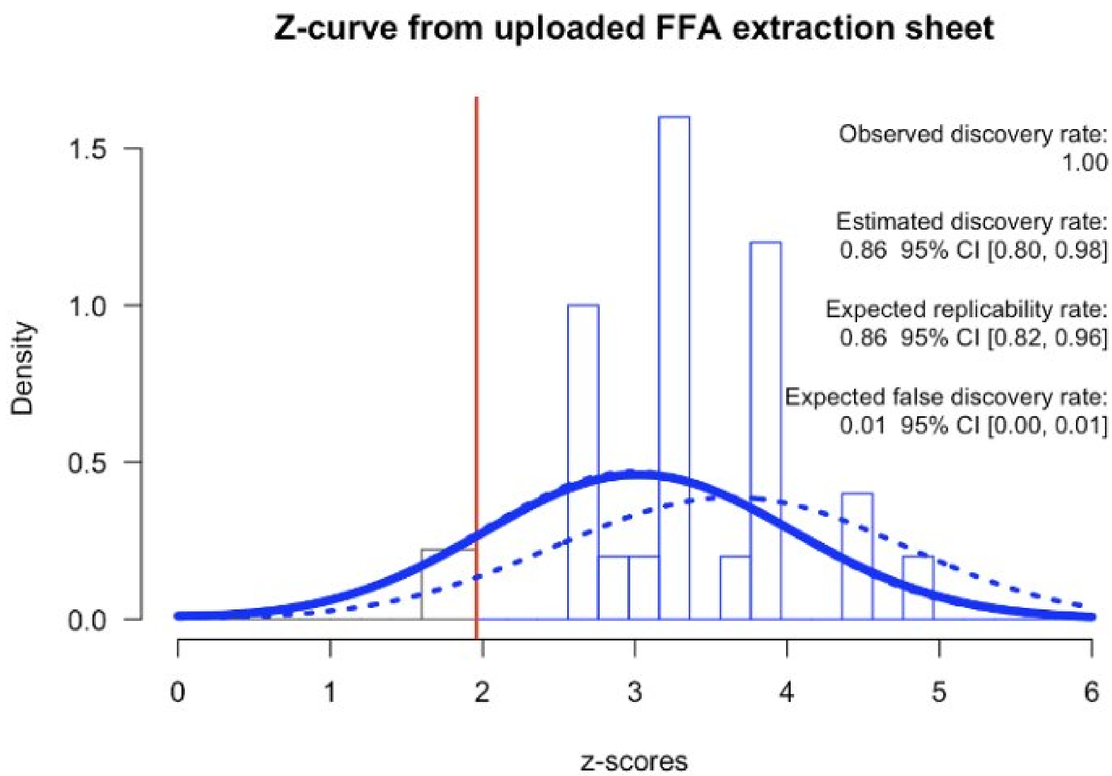

Figure 1

Z-Curve 2.0 fit to the FFA face-selectivity literature (n = 25 significant focal z-scores). The red vertical line marks the significance threshold ($z = 1.96$). The solid blue curve represents the Z-curve 2.0 model fit to the distribution of significant z-values; the dashed blue curve extrapolates the implied population density into the non-significant region, used to estimate the EDR.